# Your AI Should Never Read the Document You Wrote

*4-Layer Distillation Architecture for AI-Assisted Infrastructure Operations*

**23,854** lines → **1,254** read once → **~200** per step → **~40** at install

## Layer 1

### Work Plans

36 files | ~660 lines/file

**For humans:**

WHY decisions
Tradeoff analysis
Past incident history

**For AI:**

TOO VERBOSE
Critical steps buried

## Layer 2

### AI Context

6 files | read once at start

**Contains:**

Project status table
Deployment order
Execution guide

**Solves:**

COLD-START
ELIMINATED

## Layer 3

### Command Guardrails

117 files | one file per step

**Enforces:**

Numbered order
00 → 10 → 20 → 30
Validate → Execute → Verify

**Solves:**

CREATIVE COMMAND
ELIMINATED

## Layer 4

### Helm Values

30 files | referenced at install

**Locks:**

Chart + App versions
Resource sizes
Feature toggles

**Solves:**

VERSION DRIFT
ELIMINATED

**23,854 → 1,254 → ~200 → ~40** DEV: 2 weeks → 2 days **PROD: 1 week → 1 day (7×)**

# Layer 1 Work Plans

TOO VERBOSE

36 files · 23,854 lines total · AI reads all at once → fails · ~660 lines/file avg

**work-plans/21-monitoring-metrics.md**

```
# Monitoring Metrics Architecture

## 1. Background & Decision
Prometheus standalone: 7-day retention limit.
Options evaluated:
  - Thanos: requires sidecar, complex HA
  - Cortex: Kafka dependency, overkill
  - Mimir: single binary, object-storage backend, no Kafka
Selected: Mimir (operational simplicity wins).

## 2. Architecture Tradeoffs
AlertManager disabled → Grafana Unified Alerting.
Risk: Grafana downtime = no alerting.

## 3. Capacity Planning
DEV: ~212K active series (default 150K → OOM)

## 4. Past Incident (2026-01-15)
Chart auto-upgrade 82.0.1→82.0.2 caused CrashLoop.
Root cause: breaking PrometheusRule CRD change.
Fix: pin --version in ALL helm install commands.

## 5. Retention Strategy  ...+180 more lines
```

**What happens when AI reads this directly**

## ✕ THE FAILURE

Given 23,854 lines across 36 files, the AI:

1. **Skipped --version 82.0.2**
   → installed latest, triggered CrashLoop
2. **Ignored execution order**
   → ran Step 20 before Step 10
   → SA role not yet created → pod crash
3. **Generated creative Helm flags**
   → --set values not in any document
   → unreproducible deployment

**Root cause:**

Critical steps buried in noise.
The signal-to-noise ratio is too low
for bounded AI context windows.

*This layer is essential for humans, but it is NOT the input for AI.*

# Layer 2 AI Context

19:1 RATIO

6 files · 1,254 lines · AI reads once at session start · 19:1 distillation · Cold-start eliminated

## ai-context/00-project-summary.md

```
# Project Summary
> K8s 1.35 migration via weighted routing

| Component     | V1         | V2              | Status |
| K8s           | 1.32       | 1.35            |        |
| Redis→Valkey  | Redis 7.x  | Valkey 8.1.3    |        |
| Kafka         | ZooKeeper  | KRaft 3.9.0     |        |

## Deployment Order
00-prereq → 10-infra → 20-platform → 30-data

## Constraints
--version required · --set for secrets
```

## ai-context/99-execution-guide.md

```
# Execution Guide
> Critical: Follow order strictly.

## Step 1: 10-infrastructure
| Step | File               | Validation            |
| 1.1  | 11-k8s-cluster.md  | Cluster ACTIVE, Ready |
| 1.2  | 11.5-rbac-roles.md | 5 RBAC roles exist    |
| 1.5  | 15-crd-verify.md   | BLOCKING: CRD exists  |

## Common Issues
Key not found → 01-encryption.md / SA → 11.5-rbac-roles.md
```

### What distillation solves

## COLD-START ELIMINATED

AI reads 1,254 lines instead of 23,854.

**Instant project awareness:**

- Component versions & status
- Deployment order with dependencies
- Constraints (--version, --set)
- Known issues + fixes

**The distillation ratio is measurable:**

**23,854 lines → 1,254 lines**

**19 : 1 compression**

Same information density. Zero ambiguity.
AI reads it in one pass.

*AI does not need your reasoning. It needs your conclusion, your order, and your constraints.*

# Layer 3 Command Guardrails

READ & EXECUTE

117 files · ~200 lines/file · AI reads one file per step · Numbered order · No creative commands

## Numbered execution order

```
command-guardrails/
├── 00-prerequisites/
│   ├── 01-encryption-key.md
│   ├── 02-node-template.md
│   ├── 03-shared-storage.md
│   └── 04-storage-buckets.md
├── 10-infrastructure/
│   ├── 11-k8s-cluster.md
│   ├── 12-k8s-addons.md
│   └── 13-helm-kube-system.md
├── 20-platform/
│   ├── 21-monitoring-metrics.md ←
│   ├── 22-monitoring-logs.md
│   └── 23-monitoring-traces.md
├── 30-data/
│   ├── 31-valkey-cluster.md
│   └── 32-kafka-kraft.md
└── 80-rollback/
    ├── rollback-21-metrics.md
    └── rollback-31-valkey.md
```

## command-guardrails/21-monitoring-metrics.md

```
# 21-monitoring-metrics
> Ref: work-plans/21-monitoring-metrics.md
> Side Effects: Prometheus, Mimir, ns monitoring

## 1. Prometheus Stack

```bash
helm install prometheus-stack \
  prometheus-community/kube-prometheus-stack \
  --version 82.0.2 \
  -n monitoring \
  -f helm-values/prometheus-stack.yaml \
  --set grafana.adminPassword="$PASSWORD"
```

Verify: kubectl get pods -n monitoring
→ Prometheus, Grafana pods Running

## 2. Mimir

```bash
helm install mimir \
  grafana/mimir-distributed \
  --version 5.8.0 \
  -n monitoring \
  -f helm-values/mimir.yaml
```
```

## What guardrails enforce

**VERSION LOCKED**
--version 82.0.2
AI cannot choose latest

**VALUES REFERENCED**
-f helm-values/*.yaml
AI cannot generate flags

**ORDER ENFORCED**
00 → 10 → 20 → 30
AI cannot skip steps

**VERIFY AFTER EACH**
Pods Running check before proceeding

**DEV Lessons section**
in every PROD file: failures from DEV → prevent in PROD

*00 before 10 before 20: the AI follows the document structure, not its own reasoning.*

# Layer 4 Helm Values

ZERO DRIFT

30 files · ~40 lines/file · Referenced at helm install · AI references, never generates

## helm-values/prometheus-stack.yaml

```yaml
# LOCKED: Do not modify versions
# Sensitive values: --set at install time

prometheus:
  prometheusSpec:
    retention: 7d
    enableRemoteWriteReceiver: true
    storageSpec:
      volumeClaimTemplate:
        spec:
          storageClassName: ssd
          resources:
            requests: { storage: 60Gi }

grafana:
  enabled: true
  plugins: [grafana-opensearch-datasource]
  env:
    GF_UNIFIED_ALERTING_ENABLED: "true"
  # adminPassword: --set only, NEVER here

alertmanager:
  enabled: false
```

## helm-values/mimir.yaml

```yaml
mimir:
  structuredConfig:
    limits:
      compactor_blocks_retention_period:
        2160h  # 90 days
      ingestion_rate: 100000
      max_global_series_per_user:
        500000  # default 150K caused OOM
    multitenancy_enabled: false
    common:
      storage:
        backend: object-store
        object-store:
          bucket_name:
            mimir-storage-dev

minio:
  enabled: false

ingester:
  replicas: 1
  persistentVolume: 10Gi
```

## The Causal Chain

What AI reads

| | |
|---|---|
| Layer 1 TOO MUCH | 23,854 |

↓ distill 19:1

| | |
|---|---|
| Layer 2 read once | 1,254 |

↓ step by step

| | |
|---|---|
| Layer 3 lines/file | ~200 |

↓ references

| | |
|---|---|
| Layer 4 lines/file | ~40 |

DEV: 2wk → 2d
PROD: 1wk → 1d (7×)

*30 files. Zero version ambiguity. The AI references these values; it does not generate them.*